

Why Another Bayesian Game?

So far, we have solved Bayesian games with small payoff matrices.

However, many real-world situations involve

- private information,
- uncertainty,
- strategic signalling,
- incomplete observations.

Goal

Understand how beliefs influence strategic decisions before moving to Signaling Games and Auctions.

Example: Entry Deterrence

Consider a market with

- Incumbent Firm
- Potential Entrant

Nature first determines the incumbent's type.

$$T = \{\text{Strong}, \text{Weak}\}$$

with

$$P(\text{Strong}) = p, \quad P(\text{Weak}) = 1 - p.$$

Only the incumbent observes its type.

The entrant only knows the prior probabilities.

Players and Strategies

Entrant

- Enter
- Stay Out

Incumbent

A strategy specifies an action for every type.

$$\sigma_I : T \rightarrow \{\text{Fight, Accommodate}\}$$

Examples

(Fight, Fight)

(Fight, Accommodate)

(Accommodate, Fight)

Payoff Structure

Strong incumbent

	F	A
E	$(-2, 2)$	$(3, 1)$
O	$(0, 4)$	$(0, 4)$

Weak incumbent

	F	A
E	$(2, -2)$	$(3, 1)$
O	$(0, 2)$	$(0, 2)$

Expected Payoff of Entering

Suppose the entrant believes

$$P(\text{Strong}) = p.$$

Expected payoff from entering is

$$U(E) = p(-2) + (1 - p)(2).$$

Therefore,

$$U(E) = 2 - 4p.$$

Remaining out gives $U(O) = 0$.

Hence,

$$\text{Enter iff } p < \frac{1}{2}$$

Python Demo: Entry Decision

```
import numpy as np

p = np.linspace(0,1,11)

for belief in p:

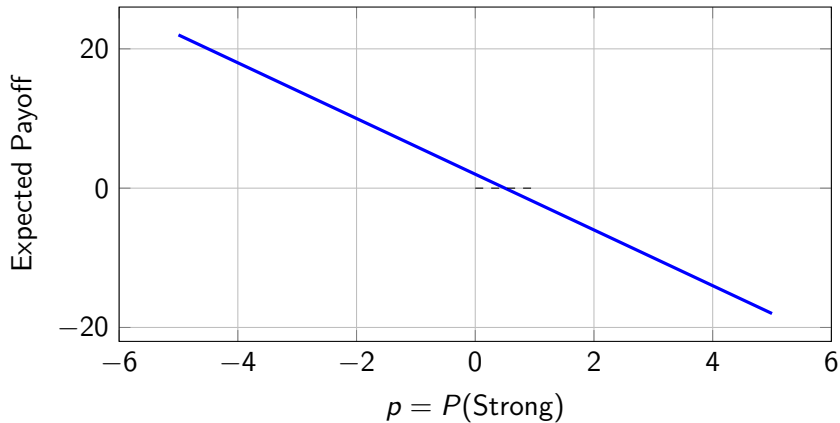
    enter = belief*(-2)+(1-belief)*2
    stay = 0

    if enter > stay:
        decision="ENTER"
    else:
        decision="STAY OUT"

    print(belief, decision)
```

Observe how the optimal strategy changes as beliefs change.

Effect of Beliefs



The intersection occurs at

$$p = \frac{1}{2}.$$

Computational Insight

Instead of solving Bayesian games manually,
we can

- enumerate strategies,
- compute expected payoffs,
- search for equilibria,
- simulate repeated play.

Modern game theory relies heavily on computation.

Examples include

- Poker AI
- AlphaGo
- Auction Design
- Security Games

Why Do We Need Signaling Games?

In Bayesian games, players possess **private information**.

Can an informed player communicate this information?

Examples:

- A job applicant knows their true ability.
- A seller knows the quality of a used car.
- A firm knows whether its product is high quality.
- A poker player knows the strength of their hand.

These situations are modeled using **Signaling Games**.

Basic Structure of a Signaling Game

A signaling game has two players.

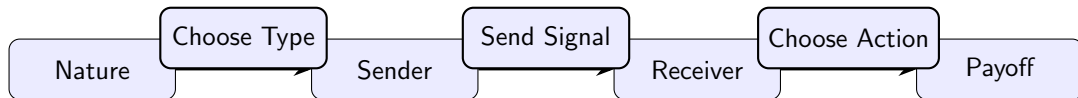
- Sender (privately informed)
- Receiver (uninformed)

Nature first chooses the Sender's type.

The Sender observes the type and sends a signal.

The Receiver observes only the signal and chooses an action.

Timeline of a Signaling Game



- Nature randomly determines the Sender's type.
- The Sender observes the type and chooses a signal.
- The Receiver observes only the signal and chooses an action.
- Payoffs depend on the type, signal, and action.

Components of a Signaling Game

A signaling game consists of

$$(N, \Theta, M, A, u_S, u_R, \mu)$$

where

- Θ : Types
- M : Signals
- A : Receiver's actions
- μ : Prior beliefs
- u_S, u_R : Utility functions

Sender's Strategy

The Sender observes

$$\theta \in \Theta.$$

A strategy is

$$\sigma_S : \Theta \rightarrow M.$$

Example

$H \rightarrow$ College Degree

$L \rightarrow$ No Degree

Receiver's Strategy

The Receiver observes only the signal.

A strategy is

$$\sigma_R : M \rightarrow A.$$

Example

Degree $\rightarrow$ Hire
No Degree $\rightarrow$ Reject

Belief Updating

The Receiver must infer the Sender's type.

Suppose

$$P(H) = 0.4, \quad P(L) = 0.6.$$

After observing a signal,
beliefs are updated using

$$P(H|m).$$

This is called the

Posterior Belief.

Bayes' Rule

Posterior beliefs satisfy

$$P(\theta|m) = \frac{P(m|\theta)P(\theta)}{P(m)}.$$

where

$$P(m) = \sum_{\theta} P(m|\theta)P(\theta).$$

Bayes' Rule connects observed signals to updated beliefs.

Bayesian Updating Example

Suppose

$$P(H) = 0.3, \quad P(L) = 0.7.$$

High-quality firms advertise with probability

0.9.

Low-quality firms advertise with probability

0.2.

What is

$$P(H|\text{Advertisement})?$$

Computing Posterior Beliefs

Using Bayes' Rule,

$$P(H|A) = \frac{0.9(0.3)}{0.9(0.3) + 0.2(0.7)} = 0.659.$$

Thus,
after observing an advertisement,
the probability that the firm is high quality increases
from

0.3

to

0.659.

Python: Bayesian Updating

```
prior = 0.3

p_signal_high = 0.9
p_signal_low  = 0.2

posterior = (p_signal_high*prior) / (
    p_signal_high*prior +
    p_signal_low*(1-prior)
)

print(posterior)
```

Changing the prior or signal reliability immediately changes the posterior belief.

Equilibrium in Signaling Games

The sender may use different signalling strategies depending on their type.

Three important outcomes can arise:

- ① **Pooling Equilibrium**
- ② **Separating Equilibrium**
- ③ **Semi-Separating (Hybrid) Equilibrium**

These equilibria differ in how much information is revealed by the sender's signal.

Pooling Equilibrium

Definition

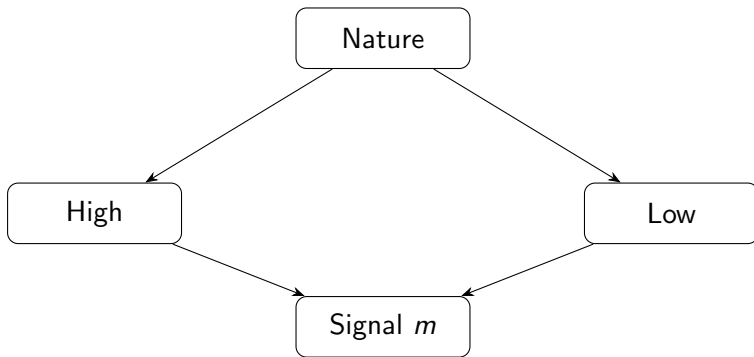
A **Pooling Equilibrium** occurs when every type of sender chooses the *same signal*.

$$\sigma(H) = \sigma(L) = m$$

Since every sender sends the same signal,

- the receiver cannot distinguish between different types;
- the signal carries no information.

Pooling Equilibrium



Both types send exactly the same signal.

Example: Pooling Equilibrium

Suppose

$$P(H) = 0.4, \quad P(L) = 0.6.$$

Both firms advertise.

$$H \rightarrow \textit{Advertise}$$

$$L \rightarrow \textit{Advertise}$$

Receiver observes only

$$\textit{Advertise}.$$

Hence

$$P(H|\textit{Advertise}) = 0.4.$$

Beliefs remain unchanged.

Properties of Pooling Equilibrium

- Sender types are indistinguishable.
- Signals reveal no information.
- Receiver relies on prior beliefs.
- Often occurs when signalling is inexpensive.

Examples

- Generic advertisements
- Uniform pricing
- Standard warranties

Separating Equilibrium

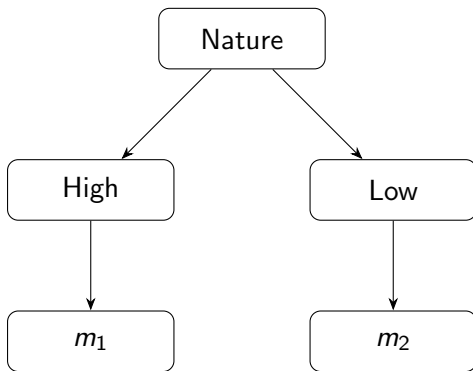
Definition

A **Separating Equilibrium** occurs when different types choose different signals.

$$\sigma(H) \neq \sigma(L)$$

The receiver can perfectly infer the sender's type from the observed signal.

Separating Equilibrium



Different types send different signals.

Example: Education Signalling

Worker types

$H, \quad L.$

Signals

$Degree, \quad No \ Degree.$

Strategies

$H \rightarrow Degree$

$L \rightarrow No \ Degree$

Employers infer the worker's type from the educational qualification.

Incentive Compatibility

For a separating equilibrium,
each type must prefer its own signal.

High type

$$u_H(m_H) \geq u_H(m_L).$$

Low type

$$u_L(m_L) \geq u_L(m_H).$$

These inequalities are called the

Incentive Compatibility Constraints.

Python: Signalling Simulation

```
types = ["High", "Low"]

signals = {
    "High": "Degree",
    "Low": "No Degree"
}

for t in types:
    print(f"{t:5s} --> {signals[t]}")
```

Modify the strategy dictionary to observe Pooling and Separating equilibria.

Semi-Separating (Hybrid) Equilibrium

Definition

A **Semi-Separating Equilibrium** occurs when

- at least one type randomizes between two or more signals,
- the receiver cannot perfectly infer the sender's type,
- posterior beliefs are updated using Bayes' Rule.

It lies between

Pooling

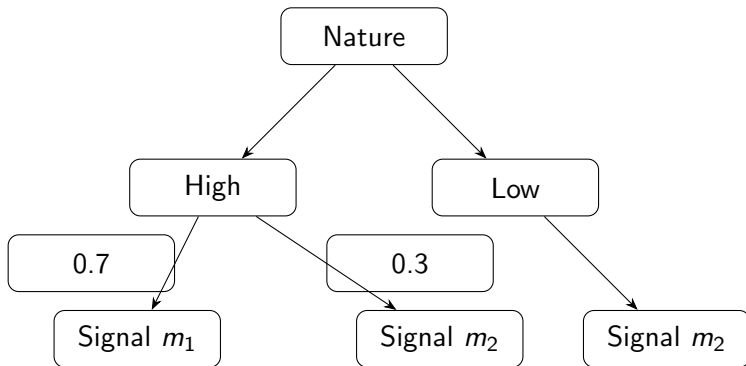


Separating

Comparing the Three Equilibria

Property	Pooling	Semi-Separating	Separating
Signal reveals information	No	Partially	Yes
Posterior beliefs	Prior	Updated	Certain
Sender randomizes	No	Yes	No
Receiver learns type	Never	Partially	Always

Illustration



High types mix between two signals while low types use one signal.

Posterior Beliefs

Suppose

$$P(H) = 0.4, \quad P(L) = 0.6.$$

High types send

$$m_1$$

with probability

$$0.7.$$

Low types send

$$m_1$$

with probability

$$0.2.$$

Updating Beliefs

Using Bayes' Rule,

$$P(H|m_1) = \frac{0.7(0.4)}{0.7(0.4) + 0.2(0.6)}.$$

Hence

$$P(H|m_1) = \frac{0.28}{0.40} = 0.70.$$

Receiving the signal

m_1

increases the belief that the sender is a high type.

Why Do Players Randomize?

Randomization occurs because

- revealing too much information may be costly,
- hiding information completely may also be costly,
- mixing signals balances these incentives.

Examples

- Poker bluffing
- Political campaigns
- Product advertising
- Salary negotiations

Python Simulation

```
import random

types = ["High","Low"]

for _ in range(10):

    t = random.choice(types)

    if t=="High":
        signal = random.choices(
            ["m1","m2"],
            [0.7,0.3])[0]
    else:
        signal = "m2"

    print(t, signal)
```

Key Takeaways

Signaling Equilibria

- Pooling: All types send the same signal.
- Separating: Each type sends a different signal.
- Semi-Separating: At least one type randomizes between signals.

The receiver updates beliefs using Bayes' Rule and chooses an optimal response.

Spence's Job Market Signaling Model

Michael Spence (1973) proposed one of the most influential signaling models in economics.

Central Question

Can education serve as a signal of ability, even if education does not increase productivity?

The model explains why employers value educational qualifications despite not directly observing a worker's true ability.

Players and Information

There are two players.

- ① Worker (Sender)
- ② Employer (Receiver)

Nature determines the worker's ability.

$$\theta \in \{H, L\}$$

where

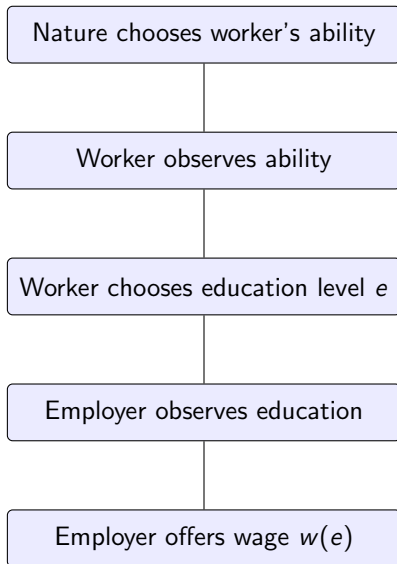
H = High Ability, L = Low Ability.

Only the worker observes the true type.

The employer knows only the prior probability

$$P(H) = p.$$

Timeline: Spence's Job Market Signaling



Strategies

The worker chooses an education level

$$e \geq 0.$$

The employer observes only

$$e$$

and decides the wage

$$w(e).$$

Thus,

$$\sigma_W : \theta \rightarrow e,$$

$$\sigma_E : e \rightarrow w.$$

Education Costs

The cost of education depends on ability.

$$c_H(e) < c_L(e),$$

for every education level e .

High-ability workers obtain education more easily.

Example

$$c_H(e) = e,$$

$$c_L(e) = 2e.$$

This difference in costs makes signaling possible.

Utilities

Worker's utility

$$U_W = w - c_\theta(e).$$

Employer's utility

$$U_E = \textit{Productivity} - \textit{Wage}.$$

Suppose

$$\textit{Productivity}(H) = 10,$$

$$\textit{Productivity}(L) = 5.$$

The employer wishes to estimate productivity using education.

Pooling Equilibrium

Suppose both worker types choose

$$e = 0.$$

The employer cannot distinguish between workers.

Therefore,

$$w = 5p + 5(1 - p).$$

Both workers receive the same wage.

Education conveys no information.

Separating Equilibrium

Suppose

$$H \rightarrow e^*$$

and

$$L \rightarrow 0.$$

The employer immediately infers high ability after observing e^* .

Hence,

$$w(e^*) = 10,$$

$$w(0) = 5.$$

Education perfectly reveals ability.

Incentive Compatibility

For separation,
each worker must prefer their own education choice.

High-ability worker

$$10 - c_H(e^*) \geq 5.$$

Low-ability worker

$$5 \geq 10 - c_L(e^*).$$

These inequalities determine the minimum education level required for a separating equilibrium.

Numerical Example

Suppose

$$c_H(e) = e,$$

$$c_L(e) = 2e.$$

Choose

$$e = 3.$$

High type: $10 - 3 = 7$

Low type: $10 - 6 = 4$

Thus, $7 \geq 5$ & $4 \leq 5$.

Only the high-ability worker chooses education.

Key Insight

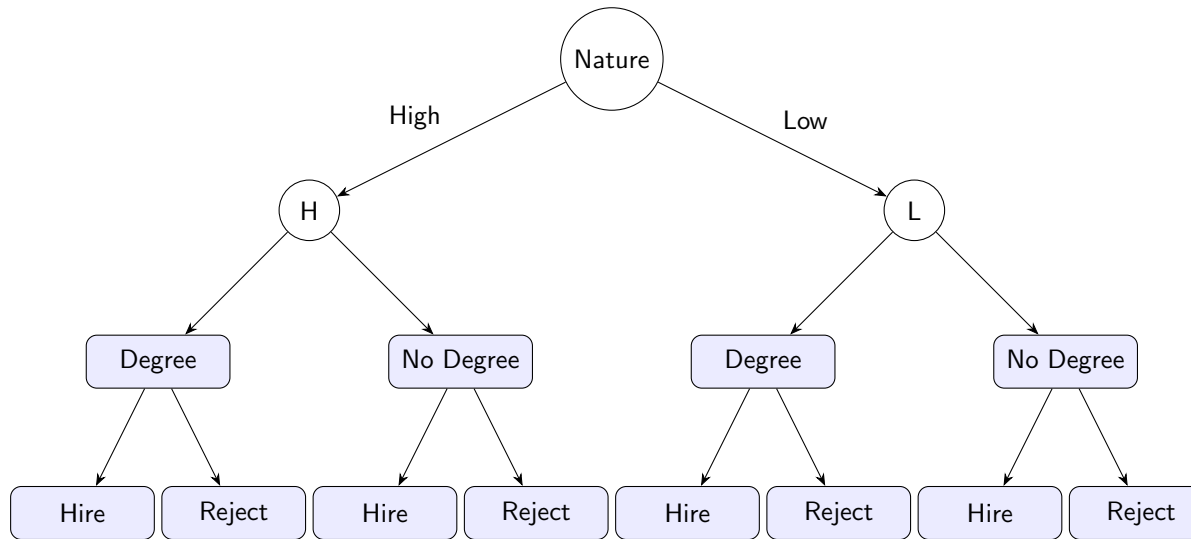
Education need not increase productivity.
Its primary role is to

Signal Ability.

The signaling mechanism works because

- education is less costly for high-ability workers,
- employers update their beliefs based on observed education,
- wages reflect inferred productivity.

Extensive Form of the Spence Signaling Game



Worker

- Observes ability
- Chooses education
- Knows own productivity

Employer

- Does not observe ability
- Observes education
- Offers wage

Observation

Education is public.

Ability is private.

Mathematical Model of the Spence Signaling Game

The game consists of the following components:

- **Worker's Type:**

$$\theta \in \Theta = \{H, L\},$$

where H and L denote High and Low ability.

- **Signal:**

$$e \in \mathbb{R}_{\geq 0},$$

where e represents the education level chosen by the worker.

- **Employer's Action:**

$$w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R},$$

where $w(e)$ is the wage offered after observing education e .

Utility Functions

Let

$$c_{\theta}(e)$$

denote the cost of acquiring education for a worker of type θ .

Worker's Utility

$$U_W(\theta, e) = w(e) - c_{\theta}(e).$$

Employer's Utility

$$U_E(\theta, e) = v(\theta) - w(e),$$

where

$$v(H) > v(L)$$

is the worker's productivity.

Strategies

The worker's strategy specifies an education level for every type.

$$\sigma_W : \Theta \rightarrow \mathbb{R}_{\geq 0}.$$

For example,

$$\sigma_W(H) = e_H, \quad \sigma_W(L) = e_L.$$

The employer's strategy specifies a wage after observing education.

$$\sigma_E : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}.$$

Thus,

$$\sigma_E(e) = w(e).$$

Cost Structure of Education

A key assumption of the Spence model is that education is *less costly* for high-ability workers. Mathematically,

$$c_H(e) < c_L(e), \quad \forall e > 0.$$

where

$c_H(e)$ = Cost for High Ability,

$c_L(e)$ = Cost for Low Ability.

A commonly used example is

$$c_H(e) = e, \quad c_L(e) = 2e.$$

Key Observation

Education does not increase productivity. Its only role is to signal the worker's ability.

How Does the Employer Determine Wages?

The employer cannot observe the worker's ability.

Instead, the employer observes only

e ,

the education level.

Based on this observation, the employer forms beliefs

$$P(H|e), \quad P(L|e).$$

The expected productivity is

$$\mathbb{E}[v|e] = P(H|e)v_H + P(L|e)v_L.$$

In a competitive labour market,

$$w(e) = \mathbb{E}[v|e].$$

Pooling Equilibrium

Suppose both worker types choose

$$e = 0.$$

The employer cannot distinguish between the two types.
Hence,

$$P(H|0) = P(H) = p.$$

The wage becomes

$$w = pv_H + (1 - p)v_L.$$

All workers receive the same wage.

$$w_H = w_L.$$

Separating Equilibrium

Suppose

$$H \rightarrow e_H, \quad L \rightarrow e_L,$$

where

$$e_H > e_L.$$

After observing education,
the employer knows the worker's type.
Therefore,

$$P(H|e_H) = 1,$$

$$P(H|e_L) = 0.$$

Consequently,

$$w(e_H) = v_H, \quad w(e_L) = v_L.$$

Incentive Compatibility

For a separating equilibrium to exist,
each worker must prefer their own education level.

High-ability worker

$$v_H - c_H(e_H) \geq v_L - c_H(e_L).$$

Low-ability worker

$$v_L - c_L(e_L) \geq v_H - c_L(e_H).$$

Interpretation

Neither type should benefit by imitating the other.

Numerical Example

Suppose

$$v_H = 10, \quad v_L = 5.$$

and

$$c_H(e) = e, \quad c_L(e) = 2e.$$

Choose

$$e_H = 4, \quad e_L = 0.$$

Then

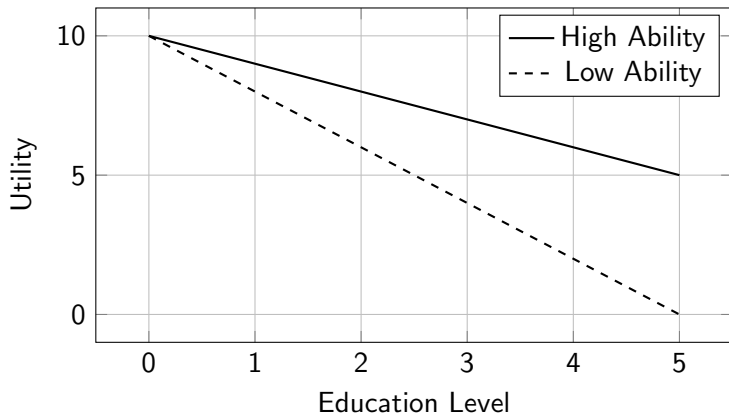
$$10 - 4 = 6 > 5,$$

so the high-ability worker prefers education.

For the low-ability worker,

$$5 - 0 = 5 < 6$$

Graphical Interpretation



The utility of high-ability workers decreases more slowly with education, making education a credible signal.

Python: Checking Incentive Compatibility

```
vH, vL = 10, 5

for e in range(6):

    high = vH - e
    low  = vH - 2*e

    print(
        f"Education = {e}",
        f"High Utility = {high}",
        f"Low Utility = {low}"
    )
```

Experiment with different cost functions to see when a separating equilibrium exists.